1. IDENTIFICATION OF THE SUBSTANCE AND THE COMPANY UNDERTAKING

1.1 Product identifier

Product name	24-5-11 TurfGro 50% XCU, 2% Fe, w WT Fe

1.2 Relevant use of the product

Applications	Fertilizers

1.3 Manufacturer, Importer or Responsible Party

Name	FERTI TECHNOLOGIES
Address	560, Chemin Rhéaume, C.P 129 J0L 2J0 Saint-Michel, Québec, Canada
Telephone	450 454-7521
Contact email	adidier@fertitechno.com

1.4 Emergency phone number

Telephone	USA National Capital Poison Center : 1 800 222 1222

2. HAZARDS IDENTIFICATION

2.1. The hazard classification of the chemical according to HCS 2012 (US-GHS)

Acute Tox. 5	H303
Skin Irrit. 2	H315
Eye Irrit. 2A	H319
STOT SE 3	H335
Carc. 1A	H350
STOT RE 1	H372

2.2. Danger symbols



2.3. Signal word Danger

2.4. Hazard statements

H303 May be harmful if swallowed
H315 Causes skin irritation
H319 Causes serious eye irritation
H335 May cause respiratory irritation
H350 May cause cancer
H372 Causes damage to organs (lungs) through prolonged or repeated exposure

2.5. Precautionary statements

Prevention

P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P260 Do not breathe dust.
P261 Avoid breathing fume/dust/gas/mist/vapors/spray.
P264 Wash hands thoroughly after handling.

		P270 Do not eat, drink or smoke when using this product. P271 Use only outdoors or in a well-ventilated area. P280 Wear protective gloves/protective clothing/eye protection/face protection.
	Response	P302+P352 IF ON SKIN: Wash with plenty of water P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing. P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P308+P313 If exposed or concerned: Get medical advice/attention. P312 Call a POISON CENTER/doctor if you feel unwell. P321 Take any precaution to avoid mixing with combustibles. P332+P313 If Skin irritation occurs get medical attention/advice. P337+P313 If eye irritation persists: get medical attention/advice. P362 Take off contaminated clothing P362 + P364 Take off contaminated clothing and wash it before reuse.
	Storage	P403+P233 Store in a well ventilated place. Keep container tightly closed. P405 Store locked up.
	Disposal	P501 Dispose of contents/container according to local regulations.
2.6.	**Description of any hazards not otherwise classified**	Not applicable.
2.7.	**% ingredient(s) with unknown acute toxicity**	Not applicable.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name		CAS-Nr.	Concentration %
XCU	Urea (85% - 95%)	57-13-6	C = 27.9%
	Sulfur (4% - 13%)	7704-34-9	
	Polymer Coating (<5%)	Mixture	
	Non-Hazardous Dye (<0.1%)	Proprietary	
46-0-0	Urea (Carbamide, carbonyldiamide, carbamidic acid) (97.5% - 99.7%)	57-13-6	C = 23.8%
	Alkalinity, as ammonia (150 ppm max)		
	Methylenediurea (0% - 2.5%)	13547-17-6	
	Biuret (0% - 1.5%)	108-19-0	
MOP 0-0-60	Potassium Chloride (95% - 99.8%)	7447-40-7	C = 17.7%
	Sodium Chloride (0.1% - 4%)	7647-14-5	
	Calcium Sulfate (0.2% - 2.5%)	7778-18-9	
Calcium Carbonate	Limestone (80% - 100%)	1317-65-3	C = 16.9%
	Quartz (SiO_2) (20%)	14808-60-7	
MAP 11-52-0	Monobasic ammonium phosphate (70% - 90%)	7722-76-1	C = 9.6%
	Ammonium sulfate (5.5% - 7.5%)	7783-20-2	
Iron sucrate	Iron oxide (70 % - < 80 %)	1317-61-9	C = 3.7%
	Calcium sulfate (3 %- < 5 %)	7778-18-9	

	Others components below reportable levels (20 % - < 30 %)	Mixture	
Wolf Trax	Ferric Oxide (60% - 100%)	1309-37-1	C = 0.3%
	Sulfonated aromatic polymer sodium salt (1% - 5%)	Proprietary	
	Silica (0.5% - 1.5%)	7631-86-9	

4. FIRST AID MEASURES

4.1 First Aid measures after Inhalation

Following inhalation: Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Use oxygen as required, provided by a qualified operator. Get medical attention if irritation develops and persists.

4.2 First Aid measures after Skin exposure

Following skin contact: Wash off immediately with plenty of water for at least 15 minutes. Take off contaminated clothing and shoes immediately. Wash contaminated clothing before re-use. Get medical attention if irritation develops and persists.

4.3 First Aid measures after Eye exposure

Following eye contact: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention if irritation develops and persists.

4.4 First Aid measures after Ingestion

Following ingestion: Induce vomiting, but only if victim is fully conscious. Never give anything by mouth to an unconscious person. Drink 1 or 2 glasses of water. Do not give milk or alcoholic beverages. Call a physician.

4.5 Most important symptoms and effects, both acute and delayed

INHALATION	May cause respiratory irritation.
SKIN	Causes skin irritation.
EYES	Causes serious eye irritation.
INGESTION	May cause stomach distress, nausea or vomiting

4.6 Indication of any immediate medical attention and special treatment needed

Notes to physician: Treat symptomatically.

5. FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable:
Use extinguishing agent suitable for type of surrounding fire. Avoid excessive water to minimize runoff. Prevent firefighter water from entering the environment.
Small fires: Water spray, foam, dry chemical or CO2
Large fires: Water spray, fog or foam.

Unsuitable: Not applicable.

5.2 Special hazards arising from chemical or mixture during the fire

Container may rupture on heating. Cool closed containers exposed to fire with water spray. Do not allow run-off from firefighting to enter drains or water courses. Explosive reactions with oxidizing agents such as potassium chlorate and/or peroxides. In case of fire hazardous decomposition products may be produced such as:

- Ammonia
- Carbon monoxide

- Carbon dioxide (CO2)

5.3 Special Protective Precautions or equipment for firefighters

In the event of fire and/or explosion do not breathe fumes. In the case of respirable dust and/or fumes, use self-contained breathing apparatus and dust impervious protective suit.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment

Wear personal protective equipment.

6.2 Emergency procedures

Unprotected persons must be kept away.
Evacuate personnel to safe areas.
Provide adequate ventilation.
Avoid dust formation.
Avoid breathing dust.
Avoid contact with skin, eyes and clothing.

6.3 Methods and materials used for containment

Do not flush into surface water or sanitary sewer system.
Prevent further leakage or spillage if safe to do so.
Do not let product enter drains.

6.4 Clean-up procedures

Use mechanical handling equipment.
Clean contaminated surface thoroughly.
Pick up and arrange disposal without creating dust.
Use a suitable vacuum cleaner.

7. HANDLING AND STORAGE

7.1 Precautions for safe handling

Handle with care.
Wear personal protective equipment.
Use only in well-ventilated areas.
Avoid dust formation.
Provide exhaust ventilation if dust is formed.
Dust must be extracted directly at the point of origin.
Avoid breathing dust.
Avoid contact with skin, eyes and clothing.

7.2 Conditions for safe storage

Keep containers tightly closed in a dry, cool and well-ventilated place.
Containers should be protected against falling down.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Store away from incompatible substances.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 ACGIH – TWA_TLV

Component / CAS	**8H (ACGIH)**
	mg/m³
MAP CAS N°: 7722-76-1	Total Respirable: 5 mg/m³ Total Inhalable: 15 mg/m³ Dust: 35 mg/m³
Calcium Sulfate CAS N°: 7778-18-9	Total Inhalable : 10 mg/m³
Sodium Chloride CAS N°: 764-14-5	Total Inhalable : 10 mg/m³

Potassium Chloride CAS Nº: 7447-40-7	Total Inhalable : 10 mg/m^3
Dust	TWA: Respirable 3 mg/m^3 Inhalable 10 mg/m^3
Urea CAS N°: 57-13-6	Respirable: 5 mg/m^3
Iron Sucrate CAS Nº: 7778-18-9	Inhalable: 10 mg/m^3
Ferric Oxide CAS Nº: 1309-37-1	Respirable: 5 mg/m^3

8.2 OSHA-Permissible Exposure Limit (PEL)

Exposure limit values of the components:

Component / CAS	**TLV, 8H (OSHA, PEL)** **mg/m^3**
Quartz (SiO2) CAS N°: 14808-60-7	Total dust: 30 mg/m^3 / %SiO2+2 (OSHA Z-3) Respirable: 10 mg/m^3 / %SiO2+2 (OSHA Z-3) Respirable: 250 mppcf / %SiO2+5 (OSHA Z-3)
Limestone CAS Nº: 1317-65-3	Total dust: 15 mg/m^3 (OSHA Z-1) Respirable : 5 mg/m^3 (OSHA Z-1) Inhalable: 15 mg/m^3 (OSHA PO) Respirable dust: 5 mg/m^3 (OSHA PO)
Urea CAS N°: 57-13-6	Respirable: 5 mg/m^3
MAP CAS N°: 7722-76-1	Total Respirable: 5 mg/m^3 Total Inhalable: 15 mg/m^3 Dust: 25 mg/m^3
Iron Sucrate CAS Nº: 7778-18-9	Respirable: 5 mg/m^3 Dust: 15 mg/m^3
Ferric Oxide CAS Nº: 1309-37-1	Fume: 10 mg/m^3
Silica CAS Nº: 7631-86-9	TWA: 0.8 mg/m^3 20 mppcf

8.3 Any other exposure limit used or recommended by chemical manufacturer

Non applicable

8.4 Engineering Controls

Provide exhaust ventilation if dust is formed. Dust must be extracted directly at the point of origin. Apply technical measures to comply with the occupational exposure limits.

8.5 Personal Protective Equipment

Hand protection: Gloves

Gloves must be inspected prior to use. Replace when worn.

Eye protection: Do not wear contact lenses.

Wear as appropriate: Safety glasses with side-shields

Body protection: Long sleeved clothing

Respiratory protection: A NIOSH approved air purifying respirator with a type 95 (R or P) particulate filter may be used under conditions where airborne concentrations are expected to exceed exposure limits. Protection provided by air purifying respirators is limited (see manufacturer's respirator selection guide). Use a positive pressure air supplied respirator if there is potential for uncontrolled release, exposure levels are not known or any other circumstances where air purifying respirators may not provide adequate protection. A respiratory protection program that meets OSHA's 29 CFR 1910.134 and ANSI Z88.2 requirements must be followed if workplace conditions warrant a respirator use.

Hygiene measures: Wash hands before breaks and at the end of workday. Remove and wash contaminated clothing before re-use. Keep working clothes separately.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information of basic physical and chemical properties

Appearance (physical state, colour, etc.)	Granular, white to brown
Odour	Odourless
Odour threshold	Not applicable
pH	No data available
Melting point/freezing point;	No data available
Boiling point	Not applicable
Boiling Range	Not applicable
Flash point	No data available
Evaporation rate	Not applicable
Flammability	Not flammable
Upper/lower flammability or explosive limits	No data available
Oxidising properties	No data available
Vapour pressure	Not applicable
Vapour density	No data available
Density	62 lbs./ft^3
Solubility in water	Partially soluble
Other Solvents	No data available
Partition coefficient (n-octanol/water)	No data available
Auto ignition temperature	No data available
Decomposition temperature	No data available

Viscosity	Not applicable

10. STABILITY AND REACTIVITY

10.1 Reactivity	Reacts violently with strong oxidants, nitrites, inorganic chlorides, chlorites and perchlorates causing fire and explosion hazard.
10.2 Chemical stability	Normally stable. May gradually give off ammonia. The product is hygroscopic and will absorb water by contact with the moisture in the air.
10.3 Possibility of hazardous reactions	If heated to the point of decomposition, oxides of phosphorus, oxides of nitrogen and ammonia (NH3) may be released.
10.4 Conditions to avoid	Moisture. High temperatures. Contact with incompatible materials.
10.5 Incompatible materials	Alkaline metals, strong oxidizing agents, strong acids or bases, nitrates, hypochlorites.
10.6 Hazardous decomposition products	If heated to the point of decomposition, oxides of phosphorus, oxides of nitrogen and ammonia (NH3) may be released. Nitrogen oxides; Ammonia; Biuret; Carbon oxides Chlorine.

11. TOXICOLOGICAL INFORMATION

11.1 Measures of Toxicity

Acute toxicity:	Sulfur: LC_{50} Inhale (Rat) = 5.43 g/m^3
Skin corrosion/irritation:	Causes skin irritation
Serious eye damage/irritation:	Causes eye irritation
Respiratory or skin sensitisation:	Product: Not a respiratory sensitizer
11.2 Listed in IARC or considered carcinogen by NTP or OSHA	Quartz (SiO2) CAS N°: 14808-60-7 Group 1 (IARC), Volume 68, 100C
11.3 Further information	This product contains prismatic tremolite (e.g., cleavage fragments) as an impurity. Sufficient exposure to respirable prismatic tremolite dust may cause serious lung problems.

12. ECOLOGICAL INFORMATION

12.1 Toxicity	The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.
12.2 Persistence and degradability	No data available
12.3 Bioaccumulative potential	No data available
12.4 Mobility in soil	No data available
12.5 Other adverse effects	

13. DISPOSAL CONSIDERATIONS

13.1 Disposal methods to employ	Recover or recycle if possible. Properly characterize all waste materials. Consult federal, state/provincial and local regulations regarding the proper disposal of this material. Prevent material from entering sewers, storm drains, other unauthorized treatment drainage systems, and natural waterways. Empty containers should be taken to an approved waste handling site for recycling or disposal.
13.2 Description of appropriate disposal containers to use	No data available
13.3 Description of the physical and chemical properties that may affect disposal activities	No data available
13.4 Language discouraging sewage disposal.	No data available
13.5 Any special precautions for landfills or incineration activities	No data available

14. TRANSPORT INFORMATION

UN Number	……
UN proper shipping name	……
Transport hazard classes	……
Packing group	……
Environmental hazards	……
Guidance On transport in bulk	……
Special precautions for user	……

15. REGULATORY INFORMATION

National and/or regional regulatory information of the chemical or mixtures

Inventories:

US. Toxic Substances Control Act: No data available

OSHA Hazards: Carcinogen

Clean Air Act: This product neither contains, nor was manufactured with a Class I or Class II ODS as defined by the U.S. Clean Air Act Section 602 (40 CFR 82, Subpt. A, App.A + B). This product does not contain any hazardous air pollutants (HAP), as defined by the U.S. Clean Air Act Section 12 (40 CFR 61).

16. OTHER INFORMATION

Indications on the revision

First edition: 04/27/2016
Addition of all fields as required by regulation (US) HCS 1910.1200 [HCS 2012].
Update of the classification information and update of related sections accordingly.

Abbreviations and acronyms used

ACGIH: American conference of governmental and industrial hygienist
CAS N°.: Chemical Abstract Service Number
CFR: Code of Federal Regulations
EC50: Half maximal effective concentration
IC50: Half maximal inhibitory concentration
HCS: Hazard communication standard
LC50: Half maximal lethal concentration
LD50: Half maximal lethal dose
OSHA: Occupational safety and health administration
STOT SE: Specific target organ toxicity Single exposure
UN N°.: United Nations Number

Methods of evaluation for the classification of mixtures

The classification of the mixture was set based on the regulation (US) HCS 1910.1200 [HCS 2012].

Other information

This information is based on our present knowledge and is provided according to the relevant national regulations. This information is intended as a characterization of the product in order to provide guidance for the relevant safety issues. However, this document does not provide any warranty, expressed or implied, regarding the properties of the product.